# Notes: Bornstein and Peter (AER, 2025)

**Nonlinear Pricing and Misallocation**

## Contents

# 1 Big picture

- **Question.** How does allowing firms to charge *nonlinear prices* — menus in which the unit price falls with quantity — change the relationship between markups, misallocation, and welfare?
- **Setting.** The paper studies a general equilibrium economy with heterogeneous firms and consumers who have heterogeneous tastes for each good. Firms cannot perfectly observe each consumer's taste, but they can post quantity-dependent price schedules and therefore engage in second-degree price discrimination.
- **Core challenge.** Most macro work on markups and misallocation assumes *linear pricing*. Under linear pricing, high-markup firms are typically too small and should be subsidized. Once firms can price nonlinearly, however, markups need not summarize the relevant distortions.
- **Main theoretical result.** With nonlinear pricing, the main distortion is *within* the firm: high-taste consumers buy too much and low-taste consumers buy too little relative to the planner's allocation. Under a broad class of preferences — constant elasticity of differences (CED) — these within-firm distortions do *not* create misallocation across firms.
- **Why this matters.** Observing large firms with high markups no longer implies that they are inefficiently small. The standard mapping from markup dispersion to firm-level misallocation is therefore not robust to nonlinear pricing.
- **Main empirical strategy.** The paper documents nonlinear pricing in NielsenIQ scanner data for grocery goods, then calibrates both a nonlinear-pricing model and a linear-pricing model to the same moments.
- **Main quantitative result.** In the benchmark nonlinear-pricing calibration, welfare losses from misallocation are equivalent to **0.68%** of permanent consumption under fixed labor supply, compared with only **0.14%** in the linear-pricing calibration. The same data therefore imply much larger misallocation once one allows firms to price nonlinearly.
- **Labor-supply result.** The aggregate labor wedge is also much smaller than a linear-pricing model would suggest. With Frisch elasticity 1, the first-best allocation has labor only **6.4%** above the market equilibrium under nonlinear pricing, versus **23.5%** under linear pricing.
- **Policy result.** Taxes and subsidies that would be optimal under linear pricing are the wrong policy under nonlinear pricing. Size-dependent subsidies that would correct linear-pricing misallocation instead reduce welfare when firms actually price nonlinearly.
- **Economic takeaway.** Nonlinear pricing creates a new, quantitatively important source of misallocation across consumers of the same firm and breaks the standard link from high markups to "firms are too small."

# 2 Data and measurement (key objects)

## 2.1 Retail data and observed price schedules

- **Data source.** The empirical analysis uses *NielsenIQ Retail Scanner Data* provided through the Kilts Center at the University of Chicago.

- **Coverage.** The dataset contains average weekly product-level prices and quantities for more than 35,000 stores.
- **Sample restriction.** The paper focuses on core grocery goods: dry groceries, frozen food, and dairy.
- **Timing.** The benchmark empirical exercise uses one week in 2017 (the week of October 16).
- **Product definition.** A product is a UPC in the scanner data.
- **Product-line definition.** A *product line* is all UPCs that share the same brand and product module. For example, different package sizes of the same brand-module combination are treated as the same product line.

## 2.2 Why this dataset is useful for the paper

- The key object in the theory is a firm's **pricing schedule** as a function of quantity purchased.
- Scanner data are especially valuable because they let the authors observe **prices paid for different package sizes of the same product line**.
- This makes it possible to measure whether unit prices decline with quantity — the core reduced-form signature of nonlinear pricing.

## 2.3 Key descriptive objects

- **Number of products:** 165,053.
- **Number of product modules:** 556.
- **Number of product lines:** 41,949.
- **Share of sales in multi-size product lines:** 90.4%.
- **Share of UPCs in multi-size product lines:** 71.3%.
- **Interpretation.** Nonlinear pricing is not a niche feature of a few odd products. Most sales occur in product lines that offer multiple package sizes.

## 2.4 Reduced-form evidence on nonlinear pricing

The paper estimates

$$\ln p_{ujs} = \beta \ln q_{ujs} + \Gamma' X_{ujs} + \varepsilon_{ujs},$$

where:

- $p_{ujs}$ is the *price per unit* of product $u$ in product line $j$ at store $s$,
- $q_{ujs}$ is package size,
- $X_{ujs}$ are fixed effects.

- The first specification includes product-line and store fixed effects.
- The second specification includes product-line-by-store fixed effects.

- In both cases, the estimates imply that a **10% larger package size is associated with only about a 4% increase in total price**. Equivalently, the per-unit price falls strongly with size.
- Robustness checks show similar nonlinearity even for **dairy** products, where storage is harder, and for the **top 5% most expensive product lines**, where packaging-cost explanations are less compelling.

*Interpretation.* The descriptive regressions are not meant to prove that all price variation comes from second-degree price discrimination. Their role is to establish that *constant per-unit pricing is empirically false*, so a nonlinear-pricing model is relevant.

### 2.5 Calibration moments and empirical targets

- The paper treats each product line as a firm.
- Firm productivity is Pareto distributed with shape parameter $\theta$.
- Structural parameters are calibrated to match four key moments:
  - dispersion of package sizes within firms,
  - market concentration (top 5% sales share),
  - aggregate cost-weighted markup,
  - elasticity of firm-level markups with respect to firm size.
- The targeted aggregate markup is **1.3**.
- The targeted elasticity of firm-level markup with respect to size is **0.03**.
- In the data, the top 5% of firms account for about **73% of sales**.

### 2.6 Key unobservables and modeling choices

- **Taste heterogeneity.** For each good, a consumer's taste is either low (1) or high ($\tau > 1$), and the realization is private information to the consumer.
- **Firm heterogeneity.** Firms differ in marginal cost $c_j$.
- **Modeling simplifications.**
  - Each product line is treated as a one-product firm.
  - The division of surplus among manufacturer, wholesaler, and retailer is abstracted from.
  - The benchmark assumes the distribution of tastes does not vary across firms.
  - Consumers buy one bundle of each good rather than arbitrary combinations of package sizes.

# 3 Models and empirical specifications (all equations + interpretation)

## 3.1 Environment: households, tastes, and budget constraint

Household $i$ has utility

$$U_i = \int_0^1 \tau_{ij} u(q_{ij})\, dj - \nu \frac{l_i^{1+\phi}}{1+\phi},$$

where:

- $q_{ij}$ is consumption of good $j$,
- $l_i$ is labor supply,
- $\tau_{ij} \in \{1, \tau\}$ is the idiosyncratic taste for good $j$,
- $\tau > 1$ means "high taste,"
- a measure $\pi$ of goods is high taste for each consumer.

The household budget constraint is

$$\int_0^1 p_j(q_{ij}) q_{ij}\, dj = l_i + \Pi,$$

where $\Pi$ denotes rebated aggregate profits.

*Interpretation.* Each firm can offer a menu $p_j(q)$ that depends on purchased quantity. Taste is private information, so firms cannot directly charge different prices by consumer type; instead they screen consumers by package size.

## 3.2 Firm problem under nonlinear pricing

Each firm $j$ has marginal cost $c_j$ and chooses a menu of bundles for low- and high-taste consumers. The firm chooses quantities $q_{1j}, q_{\tau j}$ and associated total prices $p_{1j}q_{1j}$ and $p_{\tau j}q_{\tau j}$ to solve

$$\max_{q_{1j}, q_{\tau j}, p_{1j}, p_{\tau j}} \pi q_{\tau j}(p_{\tau j} - c_j) + (1-\pi) q_{1j}(p_{1j} - c_j)$$

subject to:

- **IR for low type:**
$$u(q_{1j}) - \frac{p_{1j}q_{1j}}{P} = 0,$$
- **IC for high type:**
$$\tau u(q_{\tau j}) - \frac{p_{\tau j}q_{\tau j}}{P} = \tau u(q_{1j}) - \frac{p_{1j}q_{1j}}{P}.$$

The optimal quantities satisfy

$$\tau u'(q_{\tau j}) = \frac{c_j}{P},$$

$$u'(q_{1j}) = \frac{1-\pi}{1-\tau\pi} \frac{c_j}{P}.$$

*Interpretation.*

- The high-taste allocation has the familiar "no distortion at the top" form *conditional on* the aggregate price index $P$.
- The low-taste consumer is distorted downward because the firm wants to keep the high-taste consumer from mimicking the low-type bundle.
- In partial equilibrium, the high type would be efficient. In general equilibrium, however, the price index differs from the efficient one, so even the high type ends up distorted relative to first best.

### 3.3 Efficient allocation and the planner's benchmark

In the first best, the planner equalizes marginal utilities across consumer types for a given variety:

$$\tau u'(q_{\tau j}^{FB}) = u'(q_{1j}^{FB}),$$

and across firms according to relative marginal costs:

$$\frac{\tau u'(q_{\tau j}^{FB})}{\tau u'(q_{\tau k}^{FB})} = \frac{u'(q_{1j}^{FB})}{u'(q_{1k}^{FB})} = \frac{c_j}{c_k}.$$

*Interpretation.* The efficient planner wants no screening distortion. The ratio of allocations across firms depends only on production costs, not on the firm's markup or pricing schedule.

### 3.4 Elasticity of differences and the CED assumption

Define the elasticity of differences as

$$\eta(mc_j, \tau) = \frac{\partial \log\left(q_{\tau j}^{FE} - q_{1j}^{FE}\right)}{\partial \log(mc_j)}, \qquad mc_j = \frac{c_j}{P^{FE}}.$$

Equivalently,

$$\eta(x, \tau) = \frac{\partial \log\Big((u')^{-1}(x/\tau) - (u')^{-1}(x)\Big)}{\partial \log x}.$$

**Assumption 1 (CED).** Preferences exhibit *constant elasticity of differences*: $\eta(mc_j, \tau) = \eta$ for all $\{mc_j, \tau\}$.

*Interpretation.*

- CED means that the difference between the high-type and low-type efficient bundles scales proportionally with firm productivity.
- Under CED, the firm's over-allocation to high types and under-allocation to low types scale in exactly offsetting ways across firms.
- Therefore, the economy can feature *markup heterogeneity without misallocation across firms.*

This is the central theoretical point of the paper.

### 3.5 No across-firm misallocation under nonlinear pricing

Under CED, the paper shows:

- all firms employ the efficient amount of labor when aggregate labor is held fixed,
- all firms produce the efficient total quantity,
- but each firm allocates that quantity inefficiently across its own consumers.

*Interpretation.* Large firms can charge higher markups and still not be too small. The relevant distortion is not "too little labor at high-markup firms," but "too much quantity for high-taste buyers and too little for low-taste buyers."

### 3.6 Aggregate labor wedge and the optimal uniform subsidy

When labor is elastic, the marginal product of labor is

$$MPL = \delta \frac{\tau u'(q_{\tau j})}{c_j} + (1-\delta)\frac{u'(q_{1j})}{c_j},$$

while the real wage is

$$\frac{1}{P} = \delta \frac{\tau u'(q_{\tau j})}{c_j} + (1-\delta)\frac{1-\tau\pi}{1-\pi}\frac{u'(q_{1j})}{c_j}.$$

Hence,

$$MPL - \frac{1}{P} = (1-\delta)\frac{\pi}{1-\pi}(\tau-1)\frac{u'(q_{1j})}{c_j} > 0.$$

*Interpretation.*

- The real wage is below the marginal product of labor.
- This wedge comes from the distortion against low-taste consumers, not from the aggregate markup itself.
- Because the within-firm distortion is the same across firms under CED, the planner wants a **uniform subsidy**, not firm-specific subsidies.

### 3.7 Comparison to linear pricing

Under linear pricing, firms solve

$$\max_{p_j, q_{1j}, q_{\tau j}} [\pi q_{\tau j} + (1-\pi)q_{1j}](p_j - c_j)$$

subject to

$$\tau u'(q_{\tau j}) = \frac{p_j}{P}, \qquad u'(q_{1j}) = \frac{p_j}{P}.$$

*Interpretation.*

- Both consumer types face the same marginal price, so there is **no within-firm misallocation across consumers**.

- Markup heterogeneity then maps directly into **across-firm misallocation**: high-markup firms are too small.
- A planner would subsidize the large, high-markup firms and tax smaller, low-markup firms.

So the entire policy lesson flips once pricing is allowed to be nonlinear.

### 3.8 Quantitative calibration

The paper calibrates four structural parameters,

$$\{\bar{\tau}, \theta, \eta, \beta_0\},$$

to match four moments:

- dispersion of package sizes within firms,
- concentration (top-5% sales share),
- aggregate cost-weighted markup,
- elasticity of markups with respect to firm size.

Benchmark parameter estimates are:

- $\bar{\tau} = 1.19$,
- $\theta = 3.14$,
- $\eta = 4.15$,
- $\beta_0 = 0.02$.

*Interpretation.*

- $\bar{\tau} = 1.19$ means that the highest-taste consumer has marginal utility about 19% above the lowest-taste consumer for a given quantity.
- $\theta = 3.14$ helps match a highly concentrated product market.
- $\eta = 4.15$ helps deliver the targeted 30% aggregate markup.
- $\beta_0 > 0$ generates demand elasticities that decline with quantity, which in turn produces higher markups for larger firms.

### 3.9 Historical comparison: 1980s versus 2010s

The paper also studies a counterfactual meant to mimic the 1980s by targeting:

- a lower aggregate markup of **1.15**, and
- a **17% lower** market share of the top 5% of firms.

*Interpretation.*

- Under linear pricing, lower markups and concentration imply *less* misallocation and lower welfare losses.

- Under nonlinear pricing, the same changes imply *more* misallocation across consumers because the calibrated elasticity of taste differences rises.
- So the qualitative direction of the welfare comparison over time can reverse depending on whether the model allows nonlinear pricing.

# 4 Identification and empirical credibility (what makes the quantitative exercise informative)

## 4.1 What is and is not identified

- This is a **quantitative calibration**, not a causal reduced-form design.
- The paper is not identifying treatment effects from quasi-experimental variation.
- Instead, it asks whether a structured model of nonlinear pricing can jointly rationalize observed package-size dispersion, concentration, and markup moments.

## 4.2 Why the data are informative about nonlinear pricing

- The prevalence of multi-size product lines establishes that firms often offer quantity menus.
- The package-size regressions show that larger bundles have substantially lower per-unit prices.
- These facts directly motivate the core model feature: firms screen consumers through quantity-dependent prices.

## 4.3 What pins down the structural parameters

- **Taste dispersion $\bar{\tau}$.** Identified from the dispersion of package sizes within product lines. Stronger taste heterogeneity helps rationalize why firms offer both small and large packages.
- **Pareto shape $\theta$.** Identified from the degree of concentration, especially the sales share of top firms.
- **Elasticity of differences $\eta$.** Helps determine demand curvature and therefore the aggregate markup.
- **Demand-curvature parameter $\beta_0$.** Helps match how markups covary with firm size.

## 4.4 Why CED matters for the main theorem

- The no-across-firm-misallocation result is not true for arbitrary preferences.
- It relies on CED, which makes the under- and over-allocation across consumer types scale proportionally with costs.
- The paper therefore separates a *general* message — nonlinear pricing creates within-firm distortions — from a *sharper* message under CED — those distortions need not imply firm-level misallocation.

### 4.5 Main limitations

- Treating each product line as a firm abstracts from multiproduct ownership.
- The calibration uses one retail week and therefore speaks to a cross section rather than rich dynamics.
- The baseline assumes that the taste distribution does not vary across firms.
- The benchmark also rules out imperfect substitutability across package sizes at purchase, although the appendix shows the qualitative results are robust to richer environments.

### 4.6 Relation to the literature

- Relative to the macro markup-and-misallocation literature, the paper shows that the standard link from high markups to "firm too small" is an artifact of linear pricing.
- Relative to mechanism-design and IO work on second-degree price discrimination, the contribution is to embed nonlinear pricing into a tractable general equilibrium setting with heterogeneous firms.
- Relative to quantitative micro-macro work, the paper uses detailed scanner data precisely because those data reveal within-firm price variation across quantities.

## 5 Tables and figures

### 5.1 Table 1 and Table 2: nonlinear pricing is quantitatively important

Table 1—Summary Statistics

| | |
|---|---|
| Number of products | 165,053 |
| Number of product modules | 556 |
| Number of product lines | 41,949 |
| Share of sales in multi-size product lines | 90.4% |
| Share of UPCs in multi-size product lines | 71.3% |

*Notes:* This table reports summary statistics of the dataset. Products are at the UPC level. Product line is the collection of products of the same brand sold in the same product module.

Table 2—Nonlinear Pricing in the Retail Sector

| | Price per unit (ln) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Size (ln) | −0.61 | −0.64 | −0.56 | −0.39 |
| (SE) | (0.0001) | (0.0001) | (0.0004) | (0.0003) |
| Product line and store FE | ✓ | | | |
| Product line × store FE | | ✓ | ✓ | ✓ |
| Sample | All | All | Dairy | Expensive |
| Observations (millions) | 88.3 | 69.7 | 5.0 | 3.3 |

*Notes:* This table reports the results of regression (35). The first column contains both product line and store-level fixed effects and includes about 88 million observations. The second column includes product line by store-level fixed effects and includes about 70 million observations.

**Read:**

- Table 1 shows that the empirical relevance of nonlinear pricing is huge, not marginal: 90.4% of sales occur in product lines with more than one size option.
- Table 2 shows the core reduced-form fact: within a product line, larger package sizes have significantly lower per-unit prices.
- The headline magnitude is striking: a 10% larger package costs only about 4% more in total.

- The dairy and expensive-product robustness checks are important because they weaken simple packaging-cost or storage-based explanations.
- These tables justify why the paper does not want to take linear pricing as the baseline empirical model.

## 5.2 Table 3: calibration and parameter estimates

**Read:**

- Table 3 maps a small number of structural parameters into four moments that summarize within-firm size dispersion, concentration, and markups.
- The estimated upper bound of taste heterogeneity, $\bar{\tau} = 1.19$, is economically meaningful but not extreme. The paper is not generating large distortions by assuming implausibly huge taste differences.
- The Pareto shape $\theta = 3.14$ implies substantial productivity dispersion and matches the fact that the top 5% of firms account for roughly 73% of sales.
- The estimate $\eta = 4.15$ gives the model enough curvature to match a 30% aggregate markup.
- The estimate $\beta_0 = 0.02$ implies that larger firms charge slightly higher markups, which is what the data require.
- The comparison to the linear-pricing calibration is conceptually important: both models can be forced to match the same aggregate moments, but they imply different welfare decompositions.

TABLE 3—CALIBRATED MOMENTS AND PARAMETERS

| | | Model | | | | Model | |
|---|---|---|---|---|---|---|---|
| | Parameter | Benchm. | LP | Moment | Data | Benchm. | LP |
| $\bar{\tau}$ | Highest taste | 1.19 | 1.47 | Package size disp. | 0.44 | 0.44 | 0.44 |
| $\theta$ | Pareto shape | 3.14 | 3.57 | Sales share top 5% | 0.73 | 0.73 | 0.73 |
| $\eta$ | Elasticity of diffs | 4.15 | 4.20 | Aggregate markup | 1.3 | 1.3 | 1.3 |
| $\beta_0$ | Departure from CES | 0.02 | 0.06 | Markup-size elast. | 0.03 | 0.03 | 0.03 |

*Notes:* This table reports the calibrated parameters for the benchmark model as well as the linear pricing (LP) model. The LP model is identical except for the fact that firms are restricted to charging a constant unit price. The right panel reports the data moments used for calibration as well as their model counterparts.

TABLE 4—WELFARE COSTS OF MISALLOCATION (FIXED LABOR SUPPLY)

| | Benchmark | Linear pricing | Benchmark w/LP subsidies |
|---|---|---|---|
| Welfare gains | 0.68% | 0.14% | −0.12% |

*Notes:* This table reports the welfare gains in the fixed-labor efficient allocation relative to the benchmark model (column 1), the model with linear pricing (column 2), and the benchmark model when the optimal linear pricing subsidy schedule is implemented (column 3). All welfare gains are measured in consumption equivalent terms. That is, the uniform increase in consumption that would make households indifferent between the two equilibria. These welfare gains are computed while keeping the aggregate level of labor unchanged.

## 5.3 Table 4: welfare costs of misallocation under fixed labor

**Read:**

- This is one of the paper's key quantitative tables.
- In the benchmark nonlinear-pricing model, the welfare gain from moving to the fixed-labor efficient allocation is **0.68%**.
- In the linear-pricing model, the comparable gain is only **0.14%**.
- So allowing nonlinear pricing increases measured misallocation by about a factor of five.
- The comparison shows that most of the welfare loss in the benchmark comes from *consumer-level* misallocation rather than reallocation across firms.

Table 5—First-Best and Second-Best Allocations Relative to Market Allocation

| | | Nonlinear pricing (%) | | Linear pricing (%) |
|---|---|---|---|---|
| | | FB | SB | FB and SB |
| Frisch elas. $=$ 1 | Aggregate labor | $+6.4$ | $+5.9$ | $+23.5$ |
| | Welfare gains | $+0.92$ | $+0.20$ | $+2.81$ |
| Frisch elas. $=$ 0.25 | Aggregate labor | $+1.8$ | $+1.7$ | $+6.3$ |
| | Welfare gains | $+0.75$ | $+0.06$ | $+0.88$ |

*Notes:* This table presents the difference between the first- and second-best allocations relative to the market allocation. The first two columns present the results for our benchmark model. The final column presents the results for the model with a linear pricing restriction. The first two rows use $\varphi = 1$, the last two rows use $\varphi = 4$. Under linear pricing, the first- and second-best allocations coincide. Welfare gains are in consumption equivalent units. In contrast to Table 4, the welfare gains in the table take into account changes in aggregate labor.

- The third column is a policy warning: implementing the size-dependent subsidy schedule that would be optimal under linear pricing causes a welfare *loss* of about **0.12%** when the true environment has nonlinear pricing.

## 5.4 Table 5: first-best and second-best labor allocations

**Read:**

- Table 5 shows that the labor wedge under nonlinear pricing is much smaller than a linear-pricing model would imply.
- With Frisch elasticity 1, first-best labor is **6.4%** above the market allocation and delivers a welfare gain of **0.92%**.
- A second-best planner restricted to firm-level subsidies chooses a **uniform 7.4% subsidy**, which raises labor by **5.9%** and welfare by **0.20%**.
- Under linear pricing, the same aggregate-markup target would imply labor should be **23.5%** higher and welfare **2.81%** higher.
- The low-Frisch case tells the same story with smaller levels: under nonlinear pricing the labor distortion is 1.8%, compared with 6.3% under linear pricing.
- The policy intuition is that the planner wants to expand labor, but the right tool is a uniform production subsidy rather than subsidies tilted toward high-markup firms.

## 5.5 Historical comparison: the 1980s counterfactual

**Read:**

- This exercise is conceptually important because it shows that the assumed pricing technology can even flip the sign of a welfare comparison over time.
- Under linear pricing, lower markups and lower concentration in the 1980s imply *lower* misallocation: 0.05% instead of 0.14%.
- Under nonlinear pricing, the same counterfactual implies *higher* consumer-level misallocation: **0.98%** instead of 0.68%.
- Likewise, the first-best welfare gain rises from 0.92% to **1.17%** under nonlinear pricing, whereas it falls sharply under linear pricing.

- The broad lesson is that interpreting changes in markups through a linear-pricing lens can be qualitatively misleading.

### 5.6 Alternative specifications and robustness

**Read:**

- The paper studies a two-bundle model with a continuum of tastes and finds very similar welfare costs: **0.71%** instead of 0.68%.
- This matters because real products are often sold in only a few package sizes.
- The paper also studies preferences outside CED and an oligopoly/nested-CES environment. In these extensions, the dominant distortion still comes from consumer-level misallocation, and the role for firm-level reallocation remains small.
- So the precise theorem of "no firm-level misallocation" depends on CED, but the broader quantitative message is robust.

## 6 Short summary

- **Method.** The paper embeds nonlinear pricing into a general equilibrium model with heterogeneous firms and heterogeneous consumer tastes, then calibrates the model to scanner-data moments on package sizes, concentration, and markups.
- **Theory.** Nonlinear pricing creates screening distortions within firms: high-taste consumers are oversupplied and low-taste consumers are undersupplied. Under CED preferences, these distortions do not generate misallocation across firms.
- **Quantitative result.** Consumer-level misallocation is much larger than firm-level misallocation inferred from a linear-pricing model: 0.68% versus 0.14% under fixed labor supply.
- **Policy lesson.** High markups do not imply that large firms should be subsidized. Under nonlinear pricing, the relevant policy margin is a uniform subsidy that addresses the labor wedge, not size-dependent subsidies aimed at reallocating labor toward high-markup firms.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

- The paper shows that once firms can charge nonlinear prices, the standard mapping from markups to misallocation breaks down.
- It establishes that markups are no longer sufficient statistics for across-firm inefficiency.
- The main distortion becomes within-firm misallocation across consumers, and this distortion can be quantitatively large.

### 7.2 Economic intuition

- A firm using second-degree price discrimination wants to separate consumers by willingness to pay.
- To do so, it restricts the bundle sold to low-taste consumers, which creates an inefficiency even if the firm's total output is efficient.
- Because this screening distortion operates similarly across firms under CED, reallocating production toward high-markup firms does not solve the problem.
- The aggregate labor wedge then comes from the low-type distortion, not from average markups themselves.

### 7.3 Takeaway

This paper is important because it changes the economic interpretation of markup dispersion. In standard linear-pricing models, high-markup firms look too small and deserve subsidies. In Bornstein and Peter's nonlinear-pricing environment, that conclusion is generally wrong. The empirically relevant distortions occur inside firms, between their own customers. For quantitative work on markups, concentration, and welfare, modeling the pricing schedule is therefore not a detail — it is central to the answer.